

AI Usage Note – Cloud Cost Optimizer AI Agent

AI Tools Used: ChatGPT (OpenAI), Claude (Anthropic)

How AI Assisted Development

ChatGPT: Debugged Python and Streamlit errors, fixed runtime issues, assisted with GitHub setup, deployment, and documentation.

Claude: Generated initial project structure, agent loop workflow, dashboard components, and boilerplate code.

Best Help Provided by AI

Streamlit dashboard development, agent loop design, GitHub API integration, exchange-rate API integration, SQLite integration, PDF generation, testing and debugging.

What AI Got Wrong

Indentation errors, minor UI issues, dependency-related problems, and code requiring manual validation.

Manual Fixes Performed

Corrected indentation issues, fixed package installation problems, configured Streamlit deployment, added GitHub API integration, and tested all code.

Key Prompts Used

1. Build a Cloud Cost Optimizer using Python, Streamlit, SQLite and CSV input.
2. Create an AI Agent Loop for cloud cost analysis.
3. Generate recommendations for idle and oversized cloud resources.
4. Integrate external APIs for enhanced reporting.
5. Create downloadable PDF reports from analysis results.
6. Add GitHub API integration to fetch repository statistics.
7. Debug Streamlit deployment and runtime issues.

Conclusion

AI accelerated development, debugging, documentation, and deployment. All generated code was reviewed, tested, modified, and integrated manually.